DRAM stores 16 mil bits addy line is $A_n - A_0$

Data word is $D_7 - D_0$

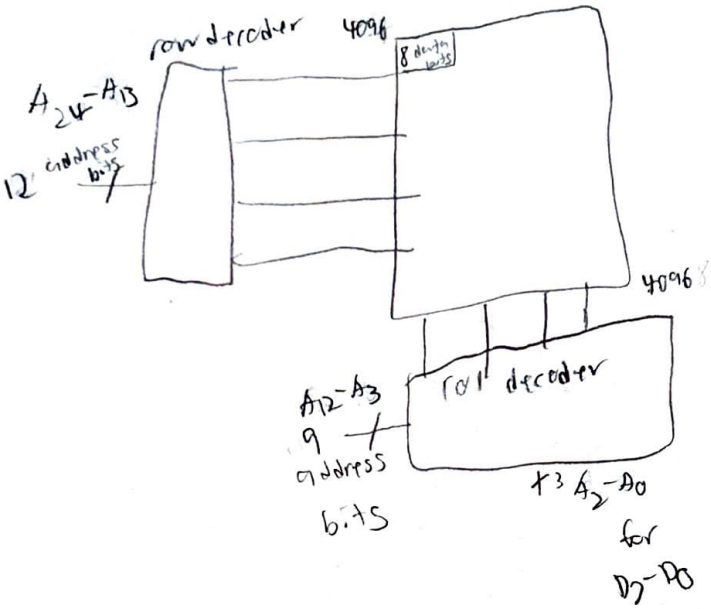
Dylan Thornburg
ECEN 153 HW9

a.

$16\ \text{mil} = 2^{24}$

$\frac{2^{24}}{8} = 2^{21}$ → 21 address $A_{20} - A_0$

$2^{21} = 2^{11} \cdot 2^{10}$



b.

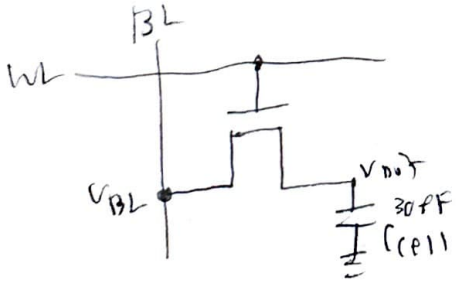
WL → BL ↓

$1\mu m(.5\mu m)(4096)(.07fF/m^2) + 2\mu m(.5\mu m)(4096)(.04fF/m^2) + 2(2\mu m)(4096)(.05fF/m) = 4.9152 \times 10^{-13} F$

$(.05fF/m)(4096) = 9.8304 \times 10^{-13} F$

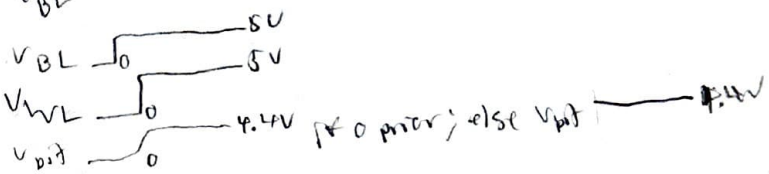
$\boxed{\text{BL cap} = 9.8304 \times 10^{-13} F}$

c.



i. write 1

$V_{BL} = '1' = 5V$, $WL = '1' = 5V$, $V_{bit} = 4.4V$

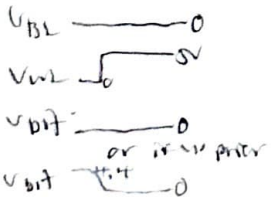


ii. write 0

$V_{BL} = '0' = 0V$

$V_{WL} = '1' = 5V$

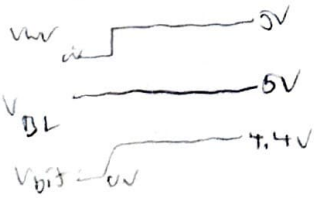
$V_{bit} = 0V$



iv. read 0

$V_{WL} = '1' = 5V$, $V_{BL} = '1' = 5V$,

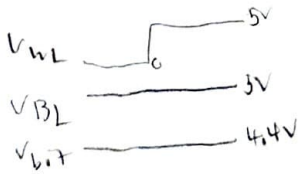
$V_{bit} = 0V$ but is charged to 4.4



must wb 0 after

ii. read 1

$V_{WL} = '1' = 5V$, $V_{BL} = '1' 5V$, $V_{bit} = 4.4V$



also wb 1 after

d.

$5(9.8304 \times 10^{-13}) = V_{new}(9.8304 \times 10^{-13} F) + 4.4(30fF)$

$V_{new} = 4.866V$

$5 - 4.866 = .134V$

Sense amp must detect a Δ of .134V